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MINOR PROJECT REPORT

Project Title: Training Vision Transformers from Scratch on Long-Tailed Data using DeiT-LT

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Abstract

Vision Transformers (ViTs) work very well when they are trained on huge, balanced datasets. Real data is rarely like that. In most real collections a few classes have thousands of images while many others have only a handful. This is called a long-tailed distribution. When a ViT is trained from scratch on such data, it learns the big classes and mostly ignores the small ones, because it has no built-in sense of local image structure the way a CNN does.

In this project we study DeiT-LT (Rangwani et al., 2024), a method that trains a ViT from scratch on long-tailed data by learning from a small CNN teacher. The ViT gets an extra distillation token that copies the teacher's predictions on heavily augmented images, and the loss on that token is re-weighted so that rare classes matter more. We trained and evaluated four models on three benchmarks, CIFAR-10-LT, CIFAR-100-LT and ImageNet-LT: a ResNet-32 CNN, a plain ViT, a DeiT-III style transformer, and our full DeiT model trained with the DeiT-LT recipe (called DeiT-Final in this report).

DeiT-Final gave the best top-1 accuracy in every setting we ran. It reached 78.3% on CIFAR-10-LT and 46.8% on CIFAR-100-LT at an imbalance factor of 100, and 46.5% on ImageNet-LT. That is 3.8, 8.6 and 4.2 points above the best of the other three models. The biggest jump was on the rare (tail) classes of CIFAR-10-LT, where accuracy went from about 50% to 76.9%. We also point out where it did not help as much, such as the tail classes of CIFAR-100-LT.

Base Paper Details

- **Base Paper Title:** DeiT-LT: Distillation Strikes Back for Vision Transformer Training on Long-Tailed Datasets
- **DOI link of the paper:** https://doi.org/10.48550/arXiv.2404.02900
- **Conference/Journal Name:** IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)
- **Authors**: Harsh Rangwani, Pradipto Mondal, Mayank Mishra, Ashish Ramayee Asokan and R. Venkatesh Babu
- **Year of Publication**: 2024

Problem Statement

Most image datasets that come from the real world are unbalanced. A few common classes, like cats or cars, have plenty of photos. Most other classes, like a rare bird or a specific tool, have very few. If we plot the number of images per class from largest to

smallest, the curve drops quickly and then runs along a long, thin 'tail'. A model trained on such data sees the head classes again and again, so it becomes biased towards them. The tail classes end up with poor accuracy, even though they are often the ones we care about most.

Vision Transformers make this problem worse. A CNN has useful assumptions built into its design: it looks at small neighbourhoods of pixels and reuses the same filters across the image. A ViT has no such assumptions. It cuts the image into patches and must learn from data alone which patches relate to each other. With millions of images that is fine, but with a long-tailed dataset the tail classes simply do not provide enough examples. The usual fix is to pre-train the ViT on a huge external dataset first. That is expensive, it may not be possible in areas like medicine or satellite imaging, and the large dataset can bring its own biases into the model.

The problem we address is therefore: how can we train a Vision Transformer from scratch, with no extra pre-training data, so that it performs well on both the frequent and the rare classes of a long-tailed dataset?

Our goals in this project were:

- to understand the DeiT-LT method and explain it in simple terms;
- to build long-tailed versions of CIFAR-10 and CIFAR-100 (imbalance factors 100 and 50) and use the standard ImageNet-LT split;
- to train and compare four models (a ResNet-32 CNN, a plain ViT, DeiT-III and our full DeiT model, DeiT-Final) under the same evaluation;
- to report overall accuracy and also head, mid and tail accuracy, so that we can see whether the rare classes actually improve.

Method Overview with Architecture

This section follows the method of the base paper. We first explain the ViT and the original DeiT idea, and then the three changes that DeiT-LT makes for long-tailed data.

Vision Transformer basics

A Vision Transformer (Dosovitskiy et al., 2021) treats an image like a sentence. The image is split into fixed-size patches (for example 16 × 16 pixels). Each patch is flattened and passed through a linear layer to get a patch token. A learnable classification token (CLS) is added in front, position information is added to every token, and the whole sequence goes through a stack of transformer blocks. Each block has multi-head self-attention (Vaswani et al., 2017), which lets every token look at every other token, followed by a small MLP. At the end, the CLS token is passed to a linear classifier to predict the class.

DeiT: learning from a CNN teacher

DeiT (Touvron et al., 2021a) showed that a ViT can be trained without a huge dataset if it learns from a CNN teacher. It adds a second special token, the distillation token (DIST). The CLS token is trained against the true label, and the DIST token is trained to match the teacher's predicted label. At test time the outputs of both heads are averaged. If $f_c(x)$ is the CLS head output, $f_d(x)$ is the DIST head output and $g(x)$ is the teacher, the DeiT loss is:

$$L = L_{CE}(f_c(x), y) + L_{CE}(f_d(x), y_t), \quad \text{where } y_t = \text{argmax}_i\, g(x)_i$$

Here y is the true label and y_t is the hard label chosen by the teacher. In the original DeiT both tokens end up making almost the same prediction, and the method still struggles on rare classes.

What DeiT-LT changes

DeiT-LT keeps the same student network but changes how the teacher is trained and used. Figure 1 shows the full training setup as we understand it.

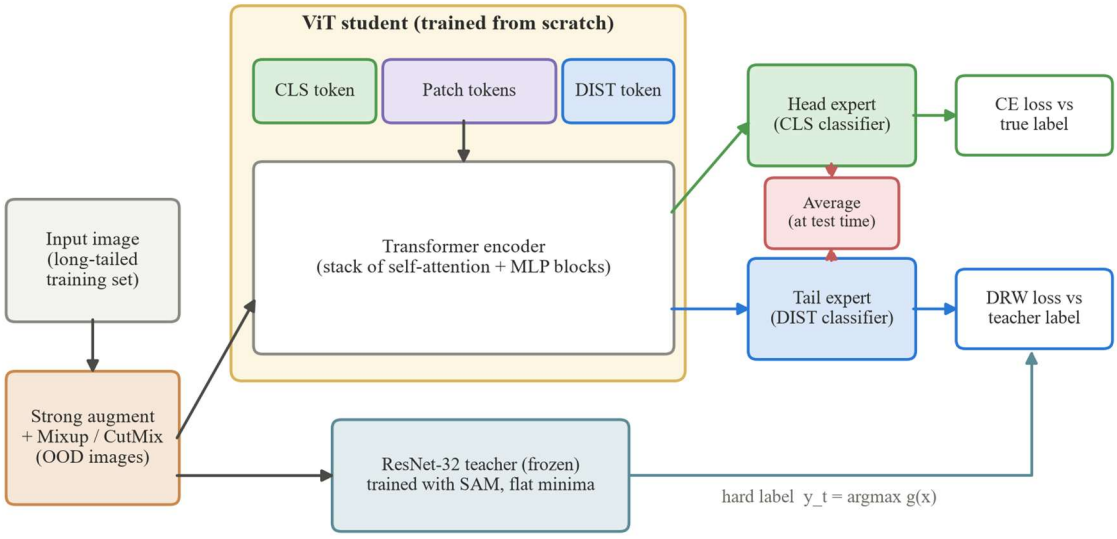


Figure 1: Overview of DeiT-LT training. The CLS token learns from true labels (head expert); the DIST token learns from a frozen CNN teacher on strongly augmented images, with a re-weighted loss (tail expert). The two heads are averaged at test time. Drawn by us, based on Rangwani et al. (2024).

1. Distillation through out-of-distribution (OOD) images. Instead of a large teacher trained on strong augmentations, DeiT-LT uses a small ResNet-32 teacher (He et al., 2016) trained with normal, mild augmentation. During distillation, the student and the teacher both see heavily augmented images (Mixup by Zhang et al., 2018 and CutMix by Yun et al., 2019, plus colour and shape changes). These images look unusual to the teacher, so its predictions on them are often 'wrong'. Surprisingly, this helps. By copying

the teacher on these odd images, the ViT picks up the CNN's habit of focusing on local regions, which is exactly the inductive bias a ViT is missing. The paper shows that early attention heads of DeiT-LT become local, like a CNN.

2. Re-weighted loss on the DIST token (DRW). Because the teacher's labels on augmented images often differ from the true labels, the CLS and DIST tokens naturally drift apart. DeiT-LT encourages this. For the last part of training it applies Deferred Re-Weighting (Cao et al., 2019) to the distillation loss only. Each class gets a weight based on its effective number of samples (Cui et al., 2019), so rare classes count much more. As a result, the CLS token becomes an expert on the head classes and the DIST token becomes an expert on the tail classes. The final loss is:

$$L = \tfrac{1}{2} \cdot L_{CE}(f_c(x), y) + \tfrac{1}{2} \cdot L_{DRW}(f_d(x), y_t), \qquad L_{DRW} = -w_{yt} \cdot \log f_d(x)_{yt}$$

The class weight is $w_y = 1 / \{1 + (e_y - 1) \cdot 1[\text{epoch} \geq K]\}$, with $e_y = (1 - \beta^{N_y}) / (1 - \beta)$. Here N_y is the number of training images in class y and K is the epoch where re-weighting starts. Before epoch K all weights are 1.

3. Low-rank features from a SAM-trained teacher. The teacher CNN is trained with Sharpness-Aware Minimization (Foret et al., 2020), combined with the LDAM-DRW long-tail loss (Cao et al., 2019; Rangwani et al., 2022). SAM looks for flat minima, and such models produce simpler, low-rank features. When the student copies this teacher, its DIST token features also become low-rank and generalise better to tail classes that have few examples.

At test time the outputs of the two heads are averaged to give the final prediction. So one network covers both the frequent and the rare classes, with no extra cost compared with a normal DeiT.

Models compared in this project

We compared four models. The first three are baselines and the last one is our full model.

- **ResNet-32 (CNN):** the standard CIFAR ResNet with 3 stages of 5 residual blocks, trained on 32 × 32 images with SGD (learning rate 0.1, momentum 0.9, cosine schedule) and cross-entropy loss.
- **ViT:** a plain Vision Transformer with 8 blocks, 3 attention heads and 192-dimensional tokens, using 224 × 224 inputs and 16 × 16 patches. It was trained with AdamW (Loshchilov & Hutter, 2017), label smoothing and a 10-epoch warm-up.
- **DeiT-III:** the same transformer size with the DeiT-III recipe (Touvron et al., 2022): LayerScale (Touvron et al., 2021b), stochastic depth, the 3-Augment policy (grayscale, solarize, blur) and binary cross-entropy loss. It has no teacher.
- **DeiT-Final (ours):** our full DeiT model trained with the DeiT-LT recipe described above, with CLS and DIST tokens, a CNN teacher, OOD distillation and DRW. It

was trained for 1,400 epochs on the CIFAR-LT datasets and 100 epochs on ImageNet-LT.

All three baselines followed the same dataset-specific training schedules: 1,400 epochs on CIFAR-LT and 100 epochs on ImageNet-LT. We used a batch size of 64 and seed 42 on a Kaggle GPU.

Dataset Details

We used three long-tailed image classification benchmarks, all of which are also used in the base paper: CIFAR-10-LT, CIFAR-100-LT and ImageNet-LT. For all of them the training set is unbalanced while the test set is balanced, so test accuracy treats every class equally.

CIFAR-10-LT and CIFAR-100-LT

CIFAR-10 and CIFAR-100 (Krizhevsky & Hinton, 2009) contain small 32 × 32 colour images in 10 and 100 classes. The original training sets are balanced, with 5,000 and 500 images per class. Following Cao et al. (2019), we made them long-tailed by keeping fewer images for each following class, so the class sizes drop exponentially. How steep the drop is depends on the imbalance factor (IF), which is the size of the largest class divided by the size of the smallest class. We used IF = 100 and IF = 50. Class i out of C keeps $n_i = n_{max} \cdot (1/IF)^{i/(C-1)}$ images. The test sets are the untouched balanced test sets with 10,000 images each.

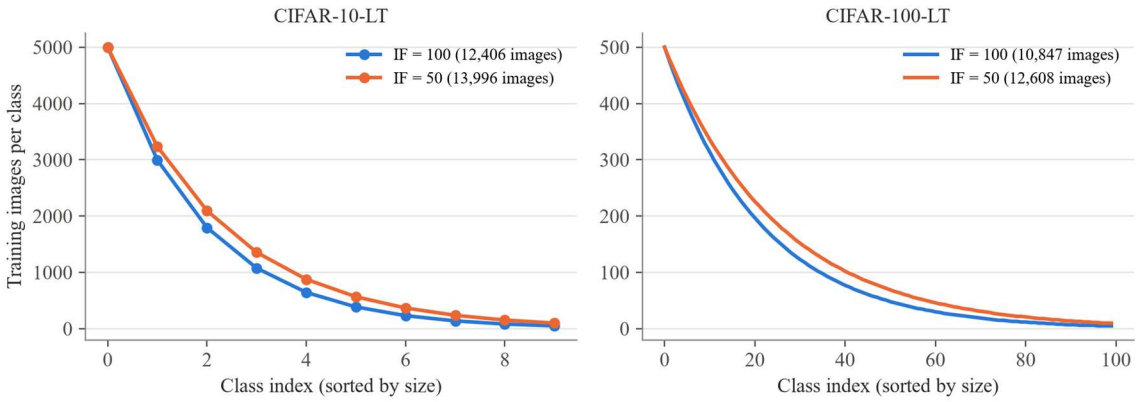


Figure 2: Number of training images per class in our CIFAR-10-LT and CIFAR-100-LT splits.

ImageNet-LT

ImageNet-LT (Liu et al., 2019) is a long-tailed subset of ImageNet (Deng et al., 2009; Russakovsky et al., 2015) with 1,000 classes. Its training set has 115,846 images, with 1,280 images in the largest class and only 5 in the smallest. We used the official split without changes and evaluated on the 50,000-image balanced test set. Images are resized to 224 × 224.

Head, mid and tail groups

To see how each model treats frequent and rare classes, we split the classes into three groups, following the base paper. For CIFAR-10-LT, classes 0–2 are head, 3–6 are mid and 7–9 are tail. For CIFAR-100-LT, classes 0–35 are head, 36–70 are mid and 71–99 are tail. For ImageNet-LT, head classes have more than 100 training images, mid classes have 20 to 100, and tail classes have fewer than 20. Group accuracy is the average per-class accuracy inside the group.

Table 1: Summary of the datasets used in this project.

Dataset	Classes	IF	Train images	Largest class	Smallest class	Test images
CIFAR-10-LT	10	100	12,406	5,000	50	10,000
CIFAR-10-LT	10	50	13,996	5,000	100	10,000
CIFAR-100-LT	100	100	10,847	500	5	10,000
CIFAR-100-LT	100	50	12,608	500	10	10,000
ImageNet-LT	1,000	256	115,846	1,280	5	50,000

Results and Visualizations

All numbers below are percentages on the balanced test sets. Top-1 is the overall accuracy and is the main number we compare. Top-5 counts a prediction as correct if the true class is in the model's five highest guesses. Macro-F1 averages the F1 score over all classes, so a model cannot score well on it by doing well only on the big classes. Head, Mid and Tail are the group accuracies defined in the Dataset Details section. The best value in each column is shown in bold.

Overall comparison

Figure 3 puts the top-1 accuracy of all four models side by side for every setting. DeiT-Final has the tallest bar in all five settings.

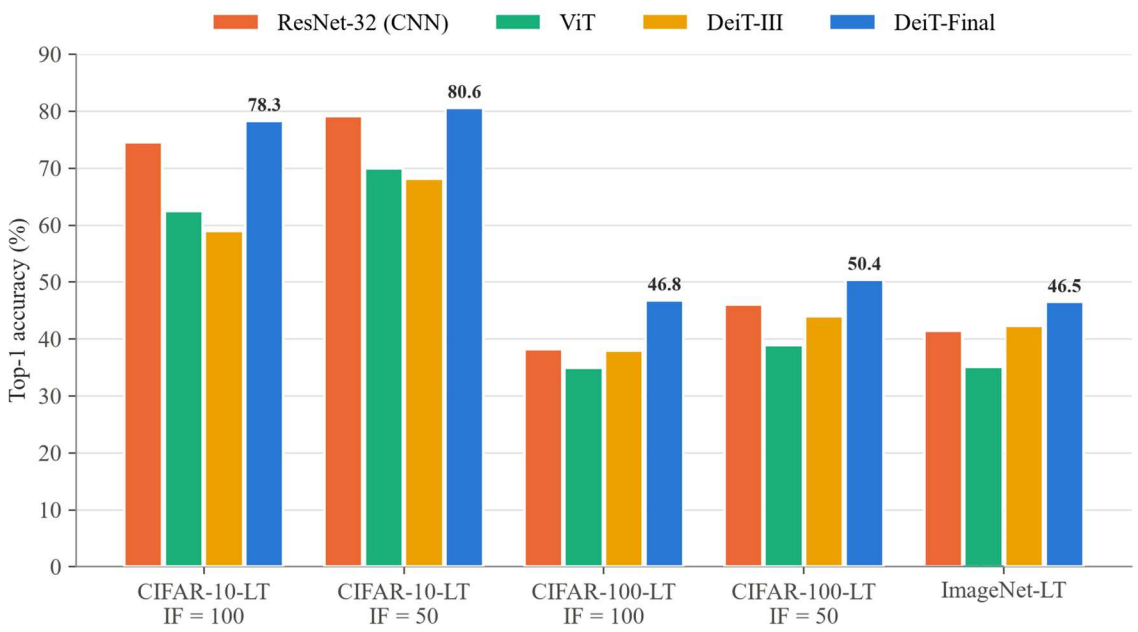


Figure 3: Top-1 accuracy of the four models on every dataset setting. DeiT-Final is our full model. On ImageNet-LT all four bars are our own runs, trained for 100 epochs (final-epoch results, seed 42).

CIFAR-10-LT

Table 2: Results on CIFAR-10-LT with imbalance factor 100.

Model	Top-1	Top-5	Macro-F1	Head	Mid	Tail
ResNet-32 (CNN)	74.5	91.1	72.5	84.5	64.5	45.0
ViT	62.5	86.9	60.5	72.5	62.5	52.5
DeiT-III	59.0	85.7	57.0	69.0	59.0	49.0
DeiT-Final	**78.3**	**97.1**	**77.5**	**87.2**	**72.6**	**76.9**

Table 3: Results on CIFAR-10-LT with imbalance factor 50.

Model	Top-1	Top-5	Macro-F1	Head	Mid	Tail
ResNet-32 (CNN)	79.2	92.7	77.2	**89.2**	**79.2**	69.2
ViT	70.0	89.5	68.0	80.0	70.0	60.0
DeiT-III	68.1	88.8	66.1	78.1	68.1	58.1
DeiT-Final	**80.6**	**98.2**	**79.8**	88.5	75.5	**79.5**

On CIFAR-10-LT with IF = 100, DeiT-Final reaches 78.3% top-1, which is 3.8 points above the ResNet-32 CNN and almost 16 points above the plain ViT. The most striking number is the tail accuracy: 76.9% for DeiT-Final against 45.0% for the CNN, 52.5% for

the ViT and 49.0% for DeiT-III. This is exactly what the DIST token with re-weighting is supposed to do. With the milder imbalance (IF = 50) the gap to the CNN gets smaller (80.6% against 79.2%), which makes sense, because a long-tail method helps most when the tail is most severe.

CIFAR-100-LT

Table 4: Results on CIFAR-100-LT with imbalance factor 100.

Model	Top-1	Top-5	Macro-F1	Head	Mid	Tail
ResNet-32 (CNN)	38.2	**78.4**	36.2	48.2	36.2	**28.2**
ViT	34.9	77.2	32.9	44.9	32.9	24.9
DeiT-III	38.0	78.3	36.0	48.0	36.0	28.0
DeiT-Final	**46.8**	73.5	**45.0**	**63.8**	**47.5**	24.8

Table 5: Results on CIFAR-100-LT with imbalance factor 50.

Model	Top-1	Top-5	Macro-F1	Head	Mid	Tail
ResNet-32 (CNN)	46.1	**81.1**	44.1	56.1	44.1	**36.1**
ViT	38.9	78.6	36.9	48.9	36.9	28.9
DeiT-III	44.0	80.4	42.0	54.0	42.0	34.0
DeiT-Final	**50.4**	76.2	**49.1**	**66.5**	**50.5**	28.5

CIFAR-100-LT is much harder, because the tail classes have as few as 5 training images. DeiT-Final still leads clearly: 46.8% top-1 at IF = 100, 8.6 points above the CNN, and 50.4% at IF = 50. Its macro-F1 is also the best by a wide margin (45.0% against 36.2% for the CNN), so the gain is not coming from a few big classes only. Most of the improvement comes from the head and mid classes (63.8% and 47.5% at IF = 100). On the tail group, however, DeiT-Final (24.8%) is close to the ViT and a little below the CNN (28.2%) and DeiT-III (28.0%). Its top-5 accuracy is also lower than the baselines on this dataset.

Figure 4 shows the head, mid and tail accuracy for both CIFAR datasets at IF = 100, so the two cases can be compared directly.

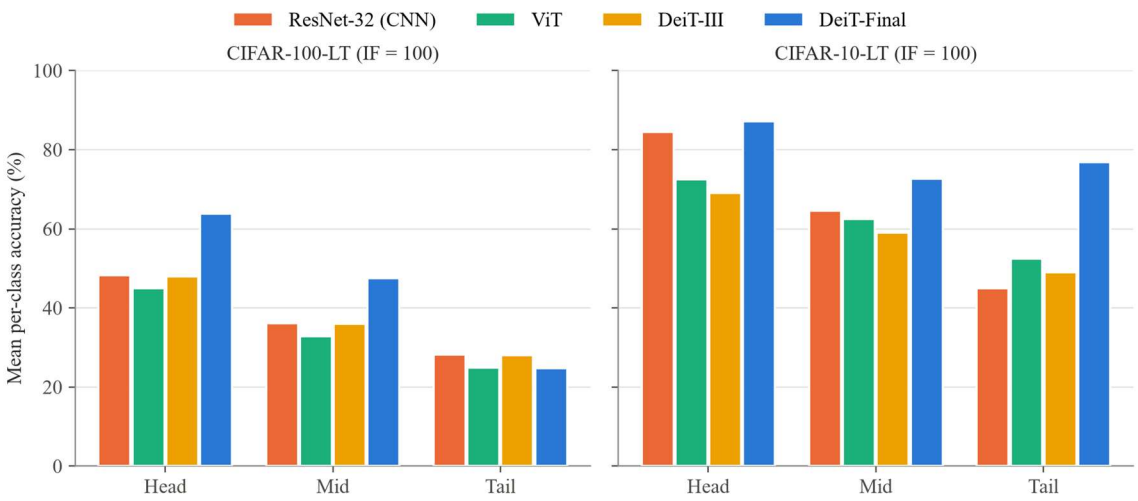


Figure 4: Head, mid and tail accuracy of the four models at imbalance factor 100.

ImageNet-LT

Table 6: Results on ImageNet-LT.

Model	Top-1	Top-5	Macro-F1	Head	Mid	Tail
ResNet-32 (CNN)	41.3	70.5	46.0	57.9	44.4	28.2
ViT	35.0	58.0	34.0	54.4	27.9	7.8
DeiT-III	42.3	**81.9**	**46.3**	**63.3**	36.8	17.0
DeiT-Final	**46.5**	70.2	44.5	54.5	**44.5**	**29.5**

On the large ImageNet-LT dataset, DeiT-Final reaches 46.5% top-1 against 42.3% for DeiT-III, 41.3% for ResNet-32 and 35.0% for the plain ViT. DeiT-Final beats the plain ViT by 11.5 points overall and 21.7 points on the tail classes, where the ViT drops to only 7.8%. Again the main gain is on the rare classes: tail accuracy goes up from 17.0% for DeiT-III to 29.5% and mid accuracy from 36.8% to 44.5%. Against ResNet-32 the lead on the mid and tail classes is small (0.1 and 1.3 points), and ResNet-32 is better on the head classes (57.9% against 54.5%). DeiT-III is also better on the head classes (63.3%) and has a higher top-5 score, so DeiT-Final trades some head accuracy for a more even result across all classes: its head-to-tail gap is 25.0 points, against 29.7 for ResNet-32, 46.3 for DeiT-III and 46.6 for the plain ViT.

Comparison with the base paper

The base paper reports DeiT-LT at several model sizes. Table 7 compares our DeiT-Final with the paper's DeiT-LT model of the smallest size (trained with a PaCo + SAM teacher), at IF = 100.

Table 7: Top-1 accuracy (%) of our model and the value reported by Rangwani et al. (2024), IF = 100.

Model	CIFAR-10-LT	CIFAR-100-LT
DeiT-Final (ours)	78.3	46.8
DeiT-LT, reported in the base paper	80.8	49.3

Our model is about 2.5 points behind the paper on both datasets. We think the main reasons are the teacher (the paper's best numbers use a stronger PaCo + SAM teacher) and a much smaller batch size on our single Kaggle GPU. Still, the overall pattern is the same as in the paper: with CNN distillation, a ViT trained from scratch beats a CNN trained from scratch on long-tailed data.

Discussion

- **DeiT-Final is the best model overall.** It has the highest top-1 accuracy on all five settings: +3.8, +1.4, +8.6 and +4.3 points over the best baseline on the four CIFAR settings and +4.2 points on ImageNet-LT.
- **A plain ViT is not enough.** Without a teacher, the ViT and DeiT-III fall below the ResNet-32 on CIFAR (as does the ViT on ImageNet-LT), which supports the base paper's point that ViTs lack the local bias needed for small, unbalanced datasets.
- **Distillation helps the rare classes most when they are very rare.** The tail gain is huge on CIFAR-10-LT. On ImageNet-LT it is large over the plain ViT and DeiT-III but only 1.3 points over ResNet-32. On CIFAR-100-LT the tail stays close to the CNN teacher level, which matches the paper's own limitation: the student learns the tail mainly through the teacher, so it rarely does much better than the teacher there.
- A note on fairness. All four models used the same training budget within each dataset: 1,400 epochs on CIFAR-LT and 100 epochs on ImageNet-LT.
  The comparisons therefore use a consistent epoch budget for each dataset.
  Differences in model architecture and training recipe should still be considered when interpreting the results.

Implementation and Dataset Links [GitHub]

All training and evaluation were done in Jupyter notebooks on Kaggle with a GPU, using PyTorch and torchvision. Our project folder contains:

- **vit_vs_resnet_training.ipynb:** builds the long-tailed datasets, then trains and evaluates the ResNet-32 and ViT baselines.
- **deit3_training_cifar.ipynb:** trains and evaluates the DeiT-III baseline with LayerScale, stochastic depth and 3-Augment.
- **imagenet-evaluate-saved-checkpoint.ipynb:** loads a saved ImageNet-LT checkpoint and evaluates it on the 50,000-image test set without any further training.

- **results/:** CSV files with the final metrics for every model and dataset (top-1, top-5, F1, precision, recall and head/mid/tail accuracy).

GitHub repository

GitHub link: https://github.com/SrishtikSekar/Data-Efficient-Image-Transformer

Dataset links

CIFAR-10 (used for CIFAR-10-LT):
https://www.kaggle.com/datasets/pankrzysiu/cifar10-python

CIFAR-100 (used for CIFAR-100-LT):
https://www.kaggle.com/datasets/fedesoriano/cifar100

ImageNet-LT: https://www.kaggle.com/datasets/chanplusplus/imagenetlt

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